KERALA SSLC | MATHEMATICS

GEOMETRY AND ALGEBRA

PYQ • 2021 - 2026

1. 2026 **4 MARKS**

(A) A circle is drawn with the line joining the points (3, 5) and (5, 9) as diameter.

(i) What are the coordinates of the centre of the circle? **(1)**

(ii) What is its radius? **(1)**

(iii) Write the equation of the circle. **(2)**

OR

(B) (i) Find the slope of the line passing through the points (3, 5) and (9, 7). **(1)**

(ii) Write the equation of the line. **(2)**

(iii) Write the coordinates of any other point on this line. **(1)**

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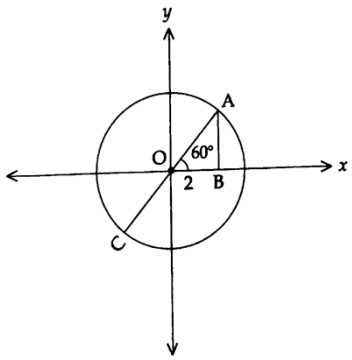
2. 2025 **5 MARKS**

In the figure, a circle is drawn with the origin as the centre. A is a point on the circle. Angle AOB = 60°; OB = 2 units.



(a) Write the coordinates of A.

(b) What is the radius of the circle?

(c) Write the equation of the circle.

(d) Write the coordinates of the point C on the circle.

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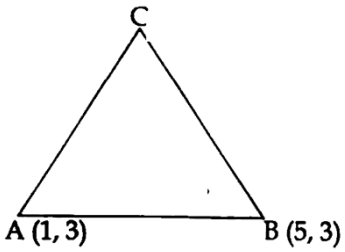
3. 2025 **4 MARKS**

In the figure, triangle ABC is an equilateral triangle. Coordinates of A are (1, 3) and coordinates of B are (5, 3).



(a) Find the length of AB.

(b) Write the coordinates of the midpoint of AB.

(c) Find the coordinates of the vertex C.

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4. 2025 **4 MARKS**

Coordinates of the points A and B are (3, 2) and (8, 7).

(a) Find the coordinates of the midpoint of the line joining the points A and B.

(b) If there is a point P with AP : PB = 2 : 3, find the coordinates of P.

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5. 2025 **3 MARKS**

A circle is drawn with (4, 3) as the centre and radius 5 units.

(a) What is the distance from the centre of the circle to the x-axis?

(b) Write the coordinates of the points where the circle cuts the x-axis.

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6. 2025 **3 MARKS**

(2, 5) and (3, 7) are two points on a line.

(a) Find the slope of the line.

(b) Write the equation of the line.

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7. 2024 **5 MARKS**

The equation of a line is $2x - y - 2 = 0$.

(a) Check whether the point (3, 4) is on this line.

(b) Find the coordinates of the points where this line cuts the x and y axes.

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8. 2024 **4 MARKS**

A(2, 3), B(8, 5) and C(4, 7) are the coordinates of the vertices of triangle ABC. P is the midpoint of AB and Q is the midpoint of BC.

(a) Find the coordinates of P and Q.

(b) Find the distance between P and Q.

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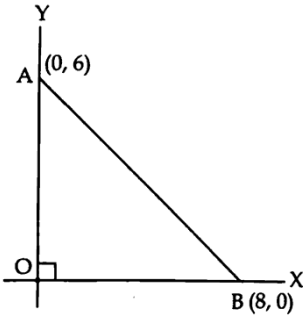
9. 2023 **5 MARKS**

In the figure, (0, 6) and (8, 0) are coordinates of the points A and B. A circle of diameter AB is to be drawn.



(a) Find the coordinates of the centre of the circle.

(b) Find the radius of the circle.

(c) What is the equation of the circle?

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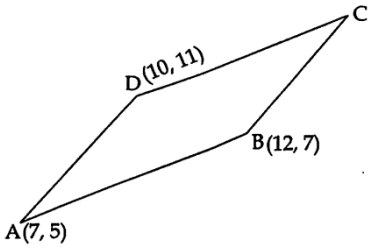
10. 2023 **4 MARKS**

In the figure, co-ordinates of 3 vertices of the parallelogram ABCD are given.



(a) Write the co-ordinates of C.

(b) Calculate the length of the diagonal AC.

(c) Find the co-ordinates of the point of intersection of the diagonals.

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11. 2023 **3 MARKS**

(4, 5) and (8, 11) are coordinates of two points on a line.

(a) Find the slope of the line.

(b) Find the equation of the line.

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12. 2022 **8 MARKS**

(a) Draw the coordinate axes and mark the points (2, 1) and (4, 3). **(3)**

(b) Find the slope of the line joining these points. **(2)**

(c) The centre of a circle is (3, 2) and the coordinates of one end of its diameter is (1, 2). Find the coordinates of the other end of the diameter. **(3)**

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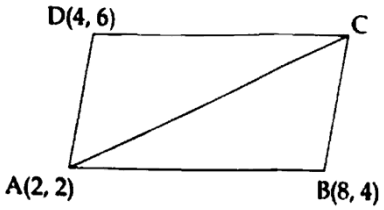
13. 2022 **4 MARKS**

The coordinates of three vertices of a parallelogram are given.



(a) Find the coordinates of the vertex C. **(2)**

(b) Find the coordinates of the midpoint of the diagonal AC. **(2)**

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14. 2022 **2 MARKS**

Find the coordinates of the point which divides the line joining the points (1, 2) and (7, 5) in the ratio 2 : 1.

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15. 2022 **1 MARK**

The equation of a line is 2x + y = 5. If the x co-ordinate of a point on this line is 2, what is the y co-ordinate of this point? (0; 1; -1; 2)

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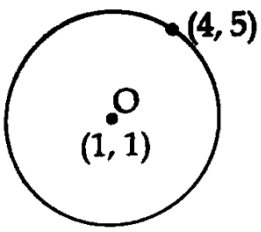
16. 2021 **5 MARKS**

A circle is drawn with (1, 1) as centre. (4, 5) is a point on the circle.



(a) Find the radius of the circle. **(1)**

(b) Write the equation of the circle. **(2)**

(c) The x coordinate of a point on the circle is 6. What is the y coordinate of that point? **(2)**

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17. 2021 **5 MARKS**

The sides of a rectangle are parallel to the axes. One pair of its opposite vertices are A(2, 4) and C(6, 12).

(a) Write the coordinates of the other two vertices. **(2)**

(b) Write the coordinates of the mid-point of AC. **(2)**

(c) x coordinate of a point on AC is 'a'. What is its y coordinate? **(1)**

18. 2021 **1 MARK**

What is the slope of the line passing through the points (2, 5) and (3, 7)? (2; 3; 4; 5)